Prescribed Performance-Based Finite-Time Consensus Technology of Nonlinear Multiagent Systems and Application to FDPs

Xinjun Wang, Ben Niu, Huanqing Wang, Xudong Zhao, and Wendi Chen

***Abstract*—This brief focuses on an adaptive bipartite consensus issue for nonlinear multi-agent systems (MASs) over signed digraphs from a new way. A new distributed control algorithm, named finite-time prescribed performance control is designed by applying prescribed performance function (PPF) and a novel first-order filter, which not only guarantees that the bipartite consensus errors converge to a prescribed compact set at a finite time, as well as enables the system to achieve the prescribed performance and fast finite time convergence. Furthermore, the neural networks (NNs) are introduced to deal with the continuous unknown nonlinearity and the effect of non-strict feedback structure that exist in the system, and the dynamic surface control (DSC) mechanism applying a novel first-order filter is used to overcome the "explosion of complexity" problem during a controller design process. Simulation experiment of forced damped pendulums (FDPs) is included to show the feasibility of the theoretical works.**

***Index Terms*—Prescribed performance, multi-agent systems, adaptive control, neural networks, forced damped pendulums.**

I. Introduction

THE RESEARCH of consensus issues of nonlinear multi-agent systems (MASs) have turned into an active research topic owing to its extensive applications in mobile satellite communication system, sensor networks, and traffic vehicle systems, e.g., [1] and the references therein. The consensus, which has attracted the research interest of a large number of scholars [2], [3], is view as the fundamental coordination problem for the study of MASs. Recently, the discuss of bipartite consensus tracking for MASs based on a signed graph has gained attention in [4], [5]. Moreover, in [6], [7], the bipartite consensus control strategy over a signed digraph is proposed for the general linear MASs. It is notable that the above results do not satisfy complex nonlinear structures. Seeing that the nonlinear dynamics exist in many branches of science and industry [8], great attention and developments have been gained in the field of complex systems control based on neural networks (NNs) [9], where NNs are regarded as an effective tool in approximating unknown nonlinear dynamics. Thus, it is very practical and urgent to research the bipartite consensus strategy of nonlinear MASs [10]–[12].

On the other hand, the most current researches are only concerned with the asymptotic stability and control problems, i.e., the system state can reach an equilibrium point in the case of time tends to infinity, which can not meet the speed of rapid convergence of the systems. Obviously, compared with the control method of infinite time convergence, the finite time method can obtain the optimal control performance in time. Recently, in [13], the bipartite consensus topic for nonlinear MASs with nonstrict feedback structure was presented by constructing a finite-time controller. Furthermore, the transient performance should be taken in to account in the practical application. For achieving the above requirements. Hence, the PPC function has been attracted the research interest of a large number of scholars in the nonlinear control field [14], [15]. Furthermore, in [8], [16], the conventional consensus problems for MASs over nonnegative graphs were solved according to PPC technology. Unfortunately, there is no significant result in distributed finite-time design issue for nonlinear MASs with prescribed performance constraints, which also motivates the research in this brief.

This brief considers finite-time control for nonlinear MASs to realize bipartite consensus with prescribed performance constraints. The main features of this brief are as follows: (1) In contrast with [8], [16], [17], a new distributed control algorithm, named prescribed performance based finite-time control technology is developed by utilizing prescribed performance function (PPF) and a novel first-order filter, which not only guarantees that the bipartite consensus errors converge to a prescribed compact set at a finite time, as well as enables the system to achieve the prescribed performance and fast finite time convergence. (2) By skillfully utilizing the property of NNs in the backstepping process, the effect of non-strict feedback structure is completely eliminated, and a novel first-order filter is designed to overcome the "explosion of complexity"

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problem. Moreover, this brief also solved the input saturation of controller, which makes the work more fit to practical applications.

II. Mathematical Description and Preliminaries

A. Graph Theory

A weighted signed digraph is described as $G \triangleq \{O, E, A\}$, where $O = \{o_1, o_2, \ldots, o_N\}$ is a group of nodes, and define the set of edges as $E \subseteq \{e_{ij} : i = 1, \ldots, N, j = 1, \ldots, N, i \neq j\}$. $A = [a_{ij}] \in R^{N\times N}$ expresses the relationship of the followers. The antagonistic interaction holds if $a_{ij} < 0$, the cooperative interaction holds when $a_{ij} > 0$. More descriptions about Graph theory can be found in [18]. It is worth indicating that a graph must include a spanning tree. G can be regarded as structurally balanced if there exists a bipartition of the sets of nodes O_1 and O_2, which satisfy $a_{ij} \geq 0$, $\forall i, j \in O_m (m = 1, 2)$; $a_{ij} < 0$, $\forall i \in O_m$, $\forall j \in O_n$, $m \neq n$ $(m, n = 1, 2)$, where $O_1 \cup O_2 = O$ and $O_1 \cap O_2 = \varnothing$.

B. Model Description

The ith agent model can be expressed as

$$\begin{aligned} \dot{x}_{i,m'} &= x_{i,m'+1} + f_{i,m'}(\bar{x}_i) + d_{i,m'}(t) \\ \dot{x}_{i,n_i} &= u_i + f_{i,n_i}(\bar{x}_i) + d_{i,n_i}(t) \\ y_i &= x_{i,1} \end{aligned} \tag{1}$$

where $m' = 1, \ldots, n_i - 1$; $\bar{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,n_i}]^T \in R^{n_i}$, $(i = 1, \ldots, N)$ is the state vector, $y_i \in R$ is the ith agent output. For $i = 1, \ldots, N, m = 1, \ldots, n_i$, $f_{i,m}(\bar{x}_i) : R^{n_i} \to R$ is unknown smooth nonlinear function. $d_{i,m}(t) \in R$ denotes interference term, which satisfies $|d_{i,m}(t)| \leq d^*_{i,m}$. $u_i \in R$ is the protocol with saturation nonlinearity [19]:

$$u_i = \text{sat}(v_i) = \begin{cases} u_{i\max}, & v_i \geq u_{i\max} \\ v_i, & u_{i\min} < v_i < u_{i\max} \\ u_{i\min}, & v_i \leq u_{i\min} \end{cases} \tag{2}$$

and $u_{i\max} > 0$, $u_{i\min} < 0$ are unknown constants. According to [19], u_i is transformed into $u_i = \varphi_{v_{\tau_i}} v_i + \bar{\varphi}(v_i)$ and $|\bar{\varphi}(v_i)| = |\text{sat}(v_i) - \varphi(v_i)| \leq \max\{u_{i\max}(1 - \tanh(1)), u_{i\min}(1 - \tanh(1))\} = \zeta_i$.

Assumption 1: Similar to [19], we assume $0 < \varrho_i \leq \varphi_{v_{\tau_i}} \leq 1$ with an unknown positive constant ϱ_i.

Lemma 1 [20]: The finite-time Lyapunov stability holds if there exists any real-values $\kappa_1 > 0$, $\kappa_2 > 0$, $0 < \eta < 1$ so that $\dot{V}(x) + \kappa_1 V(x) + \kappa_2 V^\eta(x) \leq 0$, where $T_r \leq t_0 + [1/(\kappa_1(1 - \eta))]\ln[(\kappa_1 V^{1-\eta}(t_0) + \kappa_2)/\kappa_2]$.

Lemma 2 [9]: The following property holds

$$\left\|S(\bar{\chi}_q)\right\|^2 \geq \left\|S(\bar{\chi}_p)\right\|^2 \tag{3}$$

where $S(\bar{\chi}_p) = [s_1(\bar{\chi}_p), s_2(\bar{\chi}_p), \ldots, s_l(\bar{\chi}_p)]^T$ being chosen as Gaussian function of NNs, and the node number $q \leq p$.

III. Main Results

The finite-time control technology is shown in this part for MASs (1) under the condition of prescribed performance constraints. As shown in Fig. 1, the control block diagram of this brief is given. First, we introduce a PPF, which is defined as

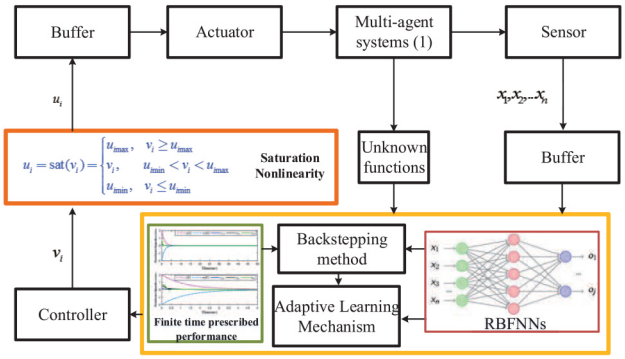


Fig. 1. The control block diagram.

$\rho_i(t) = (\rho_{i,0} - \rho_{i,\infty})e^{-c_i t} + \rho_{i,\infty}$, $\forall t \geq 0$, where $\rho_{i,0}$, $\rho_{i,\infty}$, c_i are positive constants. For $i = 1, 2, \ldots, N$, the consensus tracking error variables are established as $e_{i,1} = \sum_{j\in N_i} |a_{ij}|(x_{i,1} - \text{sign}(a_{ij})x_{j,1}) + b_i(x_{i,1} - y_0)$, which is satisfied $-\delta_{i,1}\rho_i(t) < e_{i,1}(t) < \delta_{i,2}\rho_i(t)$, where $0 < \delta_{i,1}, \delta_{i,2} \leq 1$ are the design parameters. Then, we can obtain that $e_{i,1}(t) = \rho_i(t)R(\varsigma_{i,1})$, where $R(\varsigma_{i,1}) = (\delta_{i,2}e^{\varsigma_{i,1}} - \delta_{i,1}e^{-\varsigma_{i,1}})/(e^{\varsigma_{i,1}} + e^{-\varsigma_{i,1}})$ and $\varsigma_{i,1}(t) = \frac{1}{2}\ln([e_{i,1}/\rho_i + \delta_{i,1}]/[\delta_{i,2} - e_{i,1}/\rho_i])$. For $q = 2, \ldots, n_i$, we define $e_{i,q} = x_{i,q} - \pi_{i,q}$ and $\chi_{i,q} = \pi_{i,q} - \alpha_{i,q-1}$ with $\pi_{i,q}$ being the output value of a first-order filter, $\alpha_{i,q-1}$ is virtual control protocol, $\chi_{i,q}$ is boundary error surface. In this brief, a first-order filter is constructed as

$$\varpi_{i,q}\dot{\pi}_{i,q} + \pi_{i,q} = \alpha_{i,q-1} + \text{sig}^\gamma(-\chi_{i,q}), \quad \pi_{i,q}(0) = \alpha_{i,q-1}(0) \quad \text{for } i = 1, 2, \ldots, N \text{ and } q = 2, \ldots, n_i \tag{4}$$

where $\varpi_{i,q} > 0$ is the time constant, $\text{sig}^\gamma(x) = \text{sign}(x)|x|^\gamma$ with $\gamma \geq 0$, $x \in R$.

The virtual and actual law are designed as

$$\begin{aligned} \alpha_{i,1} = \frac{1}{(d_i + b_i)}\Big(&- k_{i,1}\varsigma_{i,1} - r_{i,1}\text{sig}^\gamma(\varsigma_{i,1}) \\ &+ \sum_{j\in N_i} |a_{ij}|\text{sig}(a_{ij})x_{j,2} + \frac{\dot{\rho}_i}{\rho_i}e_{i,1}\Big) - \frac{1}{2s_{i,1}}\varsigma_{i,1}\hat{\theta}_{i,1} \end{aligned} \tag{5}$$

$$\alpha_{i,q} = -k_{i,q}e_{i,q} - r_{i,q}\text{sig}^\gamma(e_{i,q}) - \frac{1}{2s_{i,q}}e_{i,q}\hat{\theta}_{i,q} + \dot{\pi}_{i,q} \quad \text{for} \quad q = 2, \ldots, n_i - 1 \tag{6}$$

$$v_i = -k_{i,n_i}e_{i,n_i} - r_{i,n_i}\text{sig}^\gamma(e_{i,n_i}) - \frac{1}{2s_{i,n_i}}e_{i,n_i}\hat{\theta}_{i,n_i} + \dot{\pi}_{i,n_i} \tag{7}$$

where $k_{i,q}$, $p_{i,q}$ for $i = 1, \ldots, N$ and $q = 1, \ldots, n_i$ are positive constants, and $0 < \gamma < 1$. $\hat{\theta}_{i,q}$ are the adaptive laws constructed as

$$\dot{\hat{\theta}}_{i,1} = \frac{p_i(d_i + b_i)\tau_{i,1}}{2s_{i,1}}\varsigma^2_{i,1} - \tau_{i,1}\Gamma_{i,1}\hat{\theta}_{i,1} \tag{8}$$

$$\dot{\hat{\theta}}_{i,q} = \frac{\tau_{i,q}}{2s_{i,q}}e^2_{i,q} - \tau_{i,q}\Gamma_{i,q}\hat{\theta}_{i,q} \tag{9}$$

where $\tau_{i,q}$, $s_{i,q}$, $\Gamma_{i,q}$ are positive design constants. The main result can be summarized in ***Theorem 1*** according to the above algorithm design.

Theorem 1: For the nonlinear MASs (1) subject to prescribed performance constraints. If Assumption 1 holds, if the feedback controller and the adaptive laws are used

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in (7), (8), (9), then the bipartite output errors converge to a prescribed compact set at a finite-time, as well as enables the system to achieve the prescribed performance and fast finite-time convergence.

Proof: A Lyapunov function is given as

$$V = \sum_{i=1}^{N}\sum_{q=1}^{n_i} V_{i,q} = \sum_{i=1}^{N}\left(\frac{1}{2}\varsigma_{i,1}^2 + \frac{\varrho_i}{2\tau_{i,1}}\tilde{\theta}_{i,1}^2 + \sum_{q=2}^{n_i}\frac{1}{2}e_{i,q}^2 + \frac{1}{2}\chi_{i,q}^2 + \frac{\varrho_i}{2\tau_{i,q}}\tilde{\theta}_{i,q}^2\right) \tag{10}$$

The derivative of V is computed as

$$\begin{aligned}\dot{V} &= \sum_{i=1}^{N}\left(\varsigma_{i,1}\dot{\varsigma}_{i,1} + \frac{\varrho_i}{\tau_{i,1}}\tilde{\theta}_{i,1}\dot{\tilde{\theta}}_{i,1} + \sum_{q=2}^{n_i}\left(e_{i,q}\dot{e}_{i,q} + \chi_{i,q}\dot{\chi}_{i,q} + \frac{\varrho_i}{\tau_{i,q}}\tilde{\theta}_{i,q}\dot{\tilde{\theta}}_{i,q}\right)\right)\\ &= \sum_{i=1}^{N}\Bigg\{\varsigma_{i,1}\Big(-p_i k_{i,1}\varsigma_{i,1} - p_i r_{i,1}\mathrm{sig}^{\gamma}(\varsigma_{i,1}) + p_i(d_i+b_i)\\ &\quad\times\left(e_{i,2} + \chi_{i,2} + \bar{f}_{i,1}(\bar{z}_{i,1}) + \bar{d}_{i,1}(t) - \frac{1}{2s_{i,1}}\varsigma_{i,1}\hat{\theta}_{i,1}\right)\\ &\quad - \frac{\varrho_i}{\tau_{i,1}}\tilde{\theta}_{i,1}\dot{\tilde{\theta}}_{i,1} + \sum_{q=2}^{n_i-1}\Bigg[e_{i,k}\Big(-k_{i,k}e_{i,k} - r_{i,k}\mathrm{sig}^{\gamma}(e_{i,k})\\ &\quad - \frac{1}{2s_{i,k}}e_{i,k}\hat{\theta}_{i,k} + e_{i,k+1} + \chi_{i,k+1} + \bar{f}_{i,k}(\bar{z}_{i,k}) + d_{i,k}(t)\\ &\quad - 2e_{i,k}\Big) + \chi_{i,k}\dot{\chi}_{i,k} - \frac{\varrho_i}{\tau_{i,k}}\tilde{\theta}_{i,k}\dot{\tilde{\theta}}_{i,k}\Bigg]\\ &\quad + e_{i,n_i}\Big(-k_{i,n_i}\varphi_{v_{\tau_i}}e_{i,n_i} - r_{i,n_i}\varphi_{v_{\tau_i}}\mathrm{sig}^{\gamma}(e_{i,n_i})\\ &\quad - \varphi_{v_{\tau_i}}\frac{1}{2s_{i,n_i}}e_{i,n_i}\hat{\theta}_{i,n_i} + \bar{\varphi}(v_i) + \bar{f}_{i,n_i}(\bar{z}_{i,n_i}) + d_{i,n_i}(t)\\ &\quad - e_{i,n_i}\Big) + \chi_{i,n_i}\dot{\chi}_{i,n_i} - \frac{\varrho_i}{\tau_{i,n_i}}\tilde{\theta}_{i,n_i}\dot{\tilde{\theta}}_{i,n_i}\Bigg\}\end{aligned} \tag{11}$$

where $\bar{f}_{i,1}(\bar{z}_{i,1}) = f_{i,1}(\bar{x}_i) - \frac{\sum_{j\in N_i}|a_{ij}|\mathrm{sig}(a_{ij})}{d_i+b_i}f_{j,1}(\bar{x}_j) - \frac{b_i\dot{y}_0}{d_i+b_i}$, $\bar{d}_{i,1}(t) = d_{i,1}(t) - \frac{\sum_{j\in N_i}|a_{ij}|\mathrm{sig}(a_{ij})}{d_i+b_i}d_{j,1}(t)$, $\bar{f}_{i,k}(\bar{z}_{i,k}) = f_{i,k}(\bar{x}_k) + 2e_{i,k}$, $\bar{f}_{i,n_i}(\bar{z}_{i,n_i}) = f_{i,n_i}(\bar{x}_i) + \frac{3}{2}e_{i,n_i}$. Based on NNs, for any $\xi_{i,k}$, $k = 1, 2, \ldots, n_i$, it follows that $\bar{f}_{i,k}(\bar{z}_{i,k}) = \Phi_{i,k}S_{i,k}(\bar{z}_{i,k}) + \delta_{i,k}$, and $|\delta_{i,k}| \le \xi_{i,k}$. Define $\theta_{i,k} = l_{i,k}\|\Phi_{i,k}\|^2/\varrho_i$, and $l_{i,k} \ge S_{i,k}^T(\bar{z}_{i,k})S_{i,k}(\bar{z}_{i,k})$. From (5) - (9), it yields that

$$\begin{aligned}\dot{V} \le \sum_{i=1}^{N}\Bigg\{&-\big(p_i k_{i,1} - 2p_i(d_i+b_i)\big)\varsigma_{i,1}^2\\ &-\left(k_{i,2} - \frac{p_i(d_i+b_i)}{2}\right)e_{i,2}^2 - \sum_{q=3}^{n_i-1}\left(k_{i,q} - \frac{1}{2}\right)e_{i,q}^2\\ &-\left(k_{i,n_i}\varrho_i - \frac{1}{2}\right)e_{i,n_i}^2 - \left(\frac{1}{\varpi_{i,2}} - \frac{p_i(d_i+b_i)}{2} - \frac{1}{2}\right)\chi_{i,2}^2\\ &-\sum_{q=3}^{n_i}\left(\frac{1}{\varpi_{i,q}} - 1\right)\chi_{i,q}^2 + \frac{p_i(d_i+b_i)(s_{i,1}+\xi_{i,1}^2)}{2}\\ &+\frac{p_i(d_i+b_i)\bar{d}_{i,1}^{*2}}{2} + \sum_{q=2}^{n_i}\left(\frac{s_{i,q}}{2} + \frac{\xi_{i,q}^2}{2} + \frac{\bar{d}_{i,q}^{*2}}{2} + \frac{h_{i,q}^2}{2}\right)\\ &+\frac{1}{2}\zeta_i^2 - p_i r_{i,1}\big|\varsigma_{i,1}\big|^{\gamma+1} - \sum_{q=2}^{n_i-1}r_{i,q}\big|e_{i,q}\big|^{\gamma+1} - \sum_{q=2}^{n_i}\frac{\big|\chi_{i,q}\big|^{\gamma+1}}{\varpi_{i,q}}\\ &-r_{i,n_i}\varrho_i\big|e_{i,n_i}\big|^{\gamma+1} + \sum_{q=1}^{n_i}\varrho_i\Gamma_{i,q}\tilde{\theta}_{i,q}\hat{\theta}_{i,q}\Bigg\}\end{aligned} \tag{12}$$

From $\varrho_i\Gamma_{i,q}\tilde{\theta}_{i,q}\hat{\theta}_{i,q} \le -\frac{\varrho_i\Gamma_{i,q}(2o_i-1)}{2o_i}\tilde{\theta}_{i,q}^2 + \frac{\varrho_i\Gamma_{i,q}o_i}{2}\theta_{i,q}^2$, $o_i > \frac{1}{2}$, and substituting it into (12), it yields that

$$\begin{aligned}\dot{V} \le \sum_{i=1}^{N}\Bigg\{&-\big(p_i k_{i,1} - 2p_i(d_i+b_i)\big)\varsigma_{i,1}^2\\ &-\left(k_{i,2} - \frac{p_i(d_i+b_i)}{2}\right)e_{i,2}^2 - \sum_{q=3}^{n_i-1}\left(k_{i,q} - \frac{1}{2}\right)e_{i,q}^2\\ &-\left(k_{i,n_i}\varrho_i - \frac{1}{2}\right)e_{i,n_i}^2 - \left(\frac{1}{\varpi_{i,2}} - \frac{p_i(d_i+b_i)}{2} - \frac{1}{2}\right)\chi_{i,2}^2\\ &-\sum_{q=3}^{n_i}\left(\frac{1}{\varpi_{i,q}} - 1\right)\chi_{i,q}^2 + \frac{p_i(d_i+b_i)s_{i,1}}{2} + \frac{p_i(d_i+b_i)\xi_{i,1}^2}{2}\\ &+\frac{p_i(d_i+b_i)\bar{d}_{i,1}^{*2}}{2} + \sum_{q=2}^{n_i}\left(\frac{s_{i,q}}{2} + \frac{\xi_{i,q}^2}{2} + \frac{\bar{d}_{i,q}^{*2}}{2} + \frac{h_{i,q}^2}{2}\right)\\ &+\frac{1}{2}\zeta_i^2 - p_i r_{i,1}\big|\varsigma_{i,1}\big|^{\gamma+1} - \sum_{q=2}^{n_i-1}r_{i,q}\big|e_{i,q}\big|^{\gamma+1}\\ &-r_{i,n_i}\varrho_i\big|e_{i,n_i}\big|^{\gamma+1} - \sum_{q=2}^{n_i}\frac{\big|\chi_{i,q}\big|^{\gamma+1}}{\varpi_{i,q}} - \sum_{q=1}^{n_i}\left(\frac{\Upsilon_{i,q}}{\tau_{i,q}}\tilde{\theta}_{i,q}^2\right)^{\frac{\gamma+1}{2}}\\ &+\sum_{q=1}^{n_i}\left(\frac{\Upsilon_{i,q}}{\tau_{i,q}}\tilde{\theta}_{i,q}^2\right)^{\frac{\gamma+1}{2}} - \sum_{q=1}^{n_i}\frac{\Upsilon_{i,q}}{\tau_{i,q}}\tilde{\theta}_{i,q}^2\\ &+\sum_{q=1}^{n_i}\varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2\Bigg\}\end{aligned} \tag{13}$$

where $\Upsilon_{i,q} = \tau_{i,q}[(\varrho_i\Gamma_{i,q}(2o_i-1))/2o_i]$. If $(\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 \ge 1$, we have $[(\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2]^{\frac{\gamma+1}{2}} - (\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 + \varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2 \le (\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 - (\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 + \varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2 = \varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2$, and if $(\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 < 1$, we have $[(\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2]^{\frac{\gamma+1}{2}} - (\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 + \varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2 < 1 - (\Upsilon_{i,q}/\tau_{i,q})\tilde{\theta}_{i,q}^2 + \varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2 < 1 + \varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2$. Thus, it follows that

$$\dot{V} \le -\Gamma V - \Theta V^{\frac{\gamma+1}{2}} + \eta \tag{14}$$

where $\Gamma = \min_{1\le i\le N}\{2p_i k_{i,1} - 4p_i(d_i + b_i), 2k_{i,2} - p_i(d_i + b_i), 2k_{i,q} - 1, \ldots, 2k_{i,n_i}\varrho_i - 1, 2(\frac{1}{\varpi_{i,2}} - \frac{p_i(d_i+b_i)}{2} - \frac{1}{2}), 2(\frac{1}{\varpi_{i,q}} - 1), 2\Upsilon_{i,q}/\varrho_i\}$, $\Theta = \min_{1\le i\le N}\{2^{(1+\gamma)/2}p_i r_{i,1}, r_{i,q}2^{(1+\gamma)/2}, 2^{(1+\gamma)/2}r_{i,n_i}\varrho_i, 2^{(1+\gamma)/2}\frac{1}{\varpi_{i,q}}, (2\Upsilon_{i,q}/\varrho_i)^{(1+\gamma)/2}\}$, and $\eta = \sum_{i=1}^{N}\{\frac{p_i(d_i+b_i)s_{i,1}}{2} + \frac{p_i(d_i+b_i)\xi_{i,1}^2}{2} + \frac{p_i(d_i+b_i)\bar{d}_{i,1}^{*2}}{2} + \sum_{q=2}^{n_i}(\frac{s_{i,q}}{2} + \frac{\xi_{i,q}^2}{2} + \frac{\bar{d}_{i,q}^{*2}}{2} + \frac{h_{i,q}^2}{2}) + \frac{1}{2}\zeta_i^2 + \sum_{q=1}^{n_i}\varrho_i\Gamma_{i,q}o_i\theta_{i,q}^2\}$.

Define a constant $0 < \beta < 1$, then $\dot{V} \le -\Gamma V - \Theta V^{\frac{\gamma+1}{2}} + \eta$ can become

$$\dot{V} \le -\beta\Gamma V - (1-\beta)\Gamma V - \Theta V^{\frac{\gamma+1}{2}} + \eta \tag{15}$$

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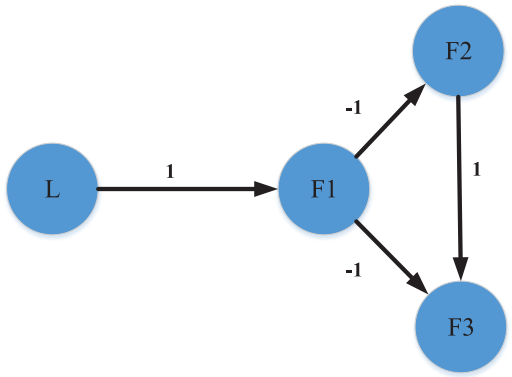


Fig. 2. Communications topology.

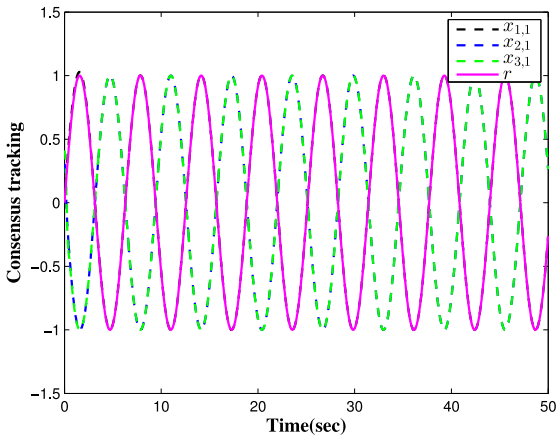


Fig. 3. Bipartite consensus performance.

or

$$\dot{V} \leq -\Gamma V - \beta\Theta V^{\frac{\gamma+1}{2}} - (1-\beta)\Theta V^{\frac{\gamma+1}{2}} + \eta \tag{16}$$

According to (15), it is easy to obtain that $\dot{V} \leq -\beta\Gamma V - \Theta V^{\frac{\gamma+1}{2}}$ when $V > [\eta/(1-\beta)\Gamma]$. Based on Lemma 1, the signals $\varsigma_{i,1}, e_{i,q}, \chi_{i,q+1}, \tilde{\theta}_{i,q}$ will converge into the region $(\varsigma_{i,1}, e_{i,q+1}, \chi_{i,q+1}, \tilde{\theta}_{i,q}) \in \{V \leq \frac{\eta}{(1-\beta)\Gamma}\}$ in a finite time $T_r \leq [1/(\beta\Gamma(1-[(1+\gamma)/2]))]\ln[(\beta\Gamma V^{1-[(1+\gamma)/2]}(0) + \Theta)/\Theta]$. Through similar analysis, according to (16), the signals $\varsigma_{i,1}, e_{i,q}, \chi_{i,q+1}, \tilde{\theta}_{i,q}$ will converge into the region $(\varsigma_{i,1}, e_{i,q+1}, \chi_{i,q+1}, \theta_{i,q}) \in \{V \leq (\frac{\eta}{(1-\beta)\Theta})^{\frac{2}{\gamma+1}}\}$ in a finite time $T_r \leq [1/(\Gamma(1-[(1+\gamma)/2]))]\ln[(\Gamma V^{1-[(1+\gamma)/2]}(0) + \beta\Theta)/\beta\Theta]$. Thus, it has $|\varsigma_{i,1}| \leq \min\{\sqrt{[2\eta/(1-\beta)\Gamma]}, \sqrt{2[\eta/(1-\beta)\Theta]^{[2/(\gamma+1)]}}\}$ in finite time $T_1 = \max\{1/(\beta\Gamma(1-[(1+\gamma)/2]))] \ln[(\beta\Gamma V^{1-[(1+\gamma)/2]}(0) + \Theta]/\Theta, 1/(\Gamma(1-[(1+\gamma)/2]))] \ln[(\Gamma V^{1-[(1+\gamma)/2]}(0) + \beta\Theta]/\beta\Theta\}$.

At this stage, the proof is completed. ■

IV. Simulation Experiment

The model of forced damped pendulums (FDPs) is described by [21]:

$$\ddot{\vartheta}_i = \frac{b_i}{m_i l_i^2}\dot{\vartheta}_i - \frac{c_i g}{l_i}\sin(\vartheta_i) + u_i, i = 1, 2, 3 \tag{17}$$

where $g = 9.8$, $l_i = 1$, $b_i = -0.25$, $c_i = 9.8^{-1}$, $\vartheta_1(0) = 0.2$, $\dot{\vartheta}_1(0) = 0.2$, $\vartheta_2(0) = 0.3$, $\dot{\vartheta}_2(0) = 0.3$, $\vartheta_3(0) = 0.4$, $\dot{\vartheta}_3(0) = 0.4$. The related parameters can be found in [21]. Let $x_{i,1} = \vartheta_i$, $x_{i,2} = \dot{\vartheta}_i$, then the system model can be rewritten as:

$$\begin{aligned} \dot{x}_{i,1} &= x_{i,2} \\ \dot{x}_{i,2} &= -\frac{b_i}{m_i l_i^2}x_{i,2} - \frac{c_i g}{l_i}\sin(x_{i,1}) + u_i \\ y_i &= x_{i,1}, i = 1, 2, 3 \end{aligned} \tag{18}$$

The leader's output is chosen as $r = 0.5(\cos t + \cos(0.5t))$. The parameters are designed as $k_{1,1} = 20$, $k_{1,2} = 10$,

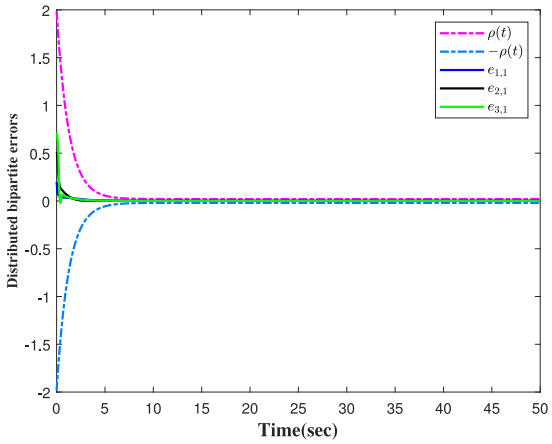


Fig. 4. Distributed bipartite errors.

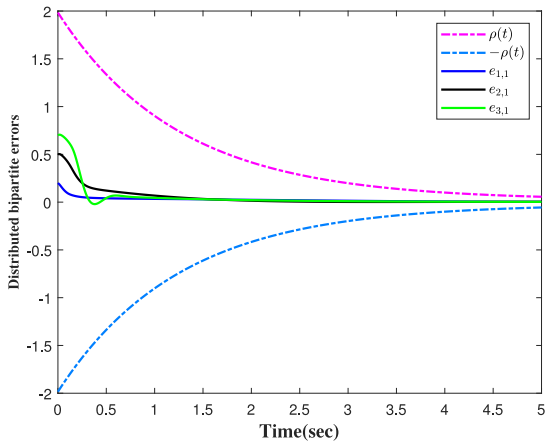


Fig. 5. Distributed bipartite errors in 5s.

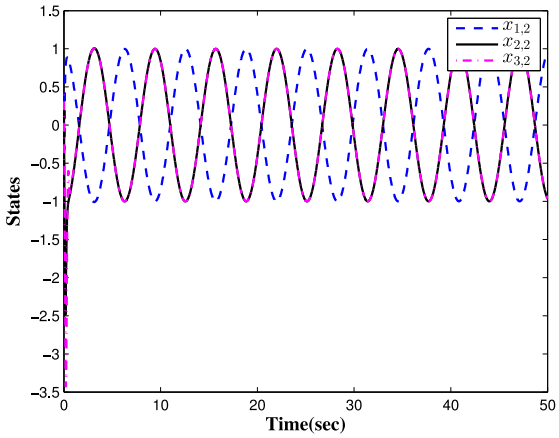


Fig. 6. States.

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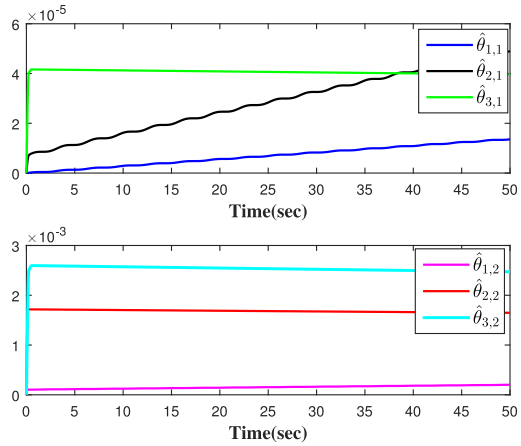


Fig. 7. The responses of adaptive laws.

$k_{2,1} = 20$, $k_{2,2} = 10$, $k_{3,1} = 20$, $k_{3,2} = 10$, $r_{1,1} = 10$, $r_{1,2} = 2$, $r_{2,1} = 2$, $r_{2,2} = 12$, $r_{3,1} = 2$, $r_{3,2} = 2$, $s_{i,q} = 1$, $\tau_{i,q} = 0.05$, $\Gamma_{i,q} = 2$, $\gamma = 0.8$. The simulation results are displayed in Figs. 3–7, respectively. Fig. 3 depicts the trajectory of the outputs of each follower and the leaders dynamics. Fig. 4–5 depict the distributed bipartite errors of each follower. Fig. 6–7 depict the curves of states and adaptive laws of each follower.

V. Conclusion

This brief has addressed the problem of finite time bipartite consensus control for nonlinear MASs over signed digraphs. By introducing PPF and a novel first-order filter, the corresponding control algorithm has been derived, which not only guarantees that the bipartite consensus errors converge to a prescribed compact set at a finite time, as well as enables the system to achieve the prescribed performance and fast finite-time convergence. Simulation experiment of forced damped pendulums (FDPs) is included to show the feasibility of the theoretical works. Furthermore, the finite time interval bipartite consensus issue is our next research work.

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